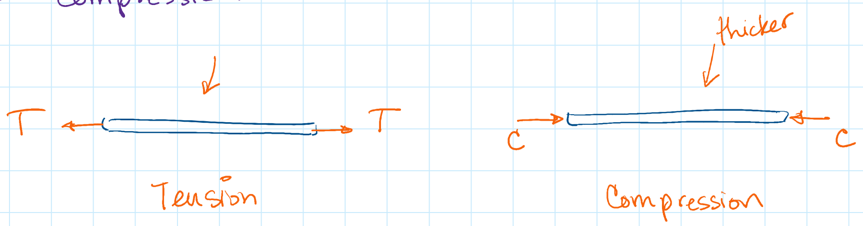
6. Structural Analysis

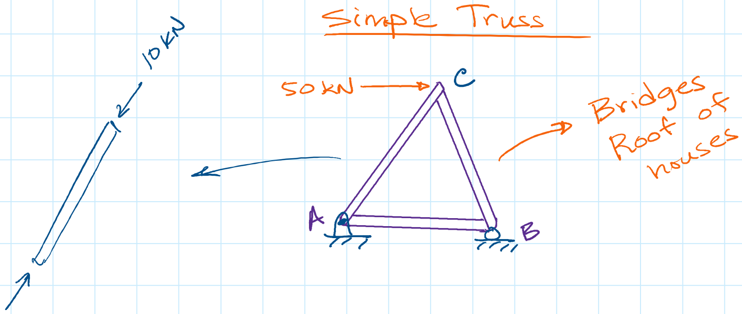
Thursday, September 26, 2024 9:42 AM

* Analyzing trusses (consists of slender members)
* Slender members can be in a state of tension or compression.



Stress - strain analysis

Simple Truss



Method of Joints

* Joints are in equilibrium
* All forces on joint must cancel each other out

$\Sigma F_x = 0 \quad \Sigma F_y = 0$

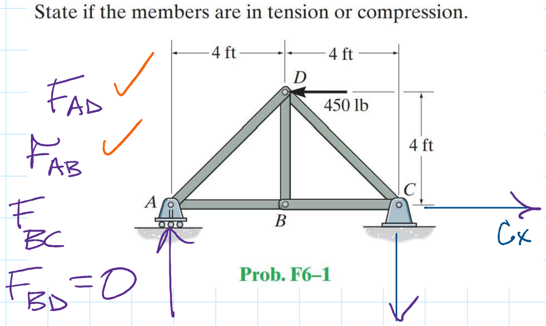
Method of Sections

* Draw a line that goes through a few members and consider a section of truss.
* Section is in equilibrium

$\Sigma F_x = 0 \quad \Sigma F_y = 0$

$\Sigma M = 0$

F6–1. Determine the force in each member of the truss. State if the members are in tension or compression.



Prob. F6–1

* Consider the truss to be uniform rigid body

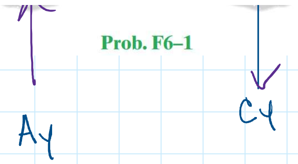
* $\Sigma F_x = 0$, $\Sigma F_y = 0$

$\Sigma M_c = 0$

* Start with ΣM_c

$F_{BD} = 0$

F_{CD}



Prob. F6–1

* Start with ΣM_C because you can neglect two unknowns (C_x & C_y). A_y is only unknown.

$$\circlearrowleft + \Sigma M_C = -8A_y + 4(450) = 0$$

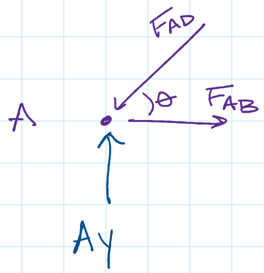
$A_y = 225$ lb $\uparrow$

$C_y = 225$ lb $\downarrow$

$C_x = 450$ lb $\rightarrow$

Method of Joints

Joint A



$\theta = 45°$

$\Sigma F_x = 0 \quad \Sigma F_y = 0$

$\Sigma F_y = A_y - F_{AD}\sin\theta = 0$

$F_{AD} = \dfrac{A_y}{\sin\theta}$ ✓

$F_{AD} = \dfrac{225}{\sin(45°)} = 318.2$ lb

$F_{AD} = 318.2$ lb (C)

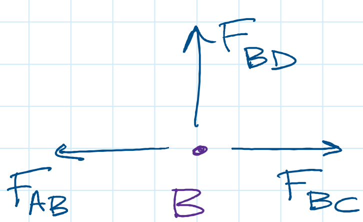
$\Sigma F_x = F_{AB} - F_{AD}\cos\theta = 0$

$F_{AB} = F_{AD}\cos\theta$

$F_{AB} = (318.2)\cos(45°) = 225$ lb

$F_{AB} = 225$ lb (T)

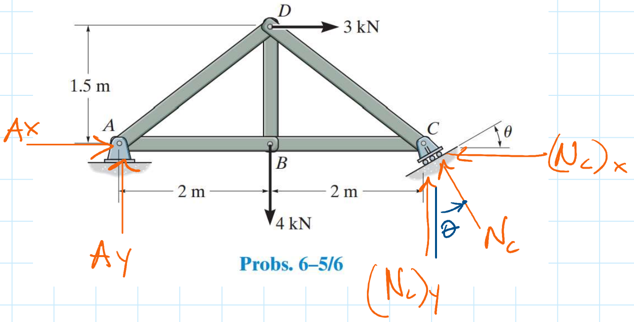
Joint B



$\Sigma F_x = 0 \quad F_{BC} - F_{AB} = 0$

$F_{BC} =$

6–6. Determine the force in each member of the truss, and state if the members are in tension or compression. Set $\theta = 30°$.



Probs. 6–5/6

$$\Sigma F_x = 0$$
$$A_x + 3 - (N_c)_x = 0$$
$$A_x = (N_c)_x - 3$$
$$A_x = N_c \sin\theta - 3$$
$$A_x = 3.6 \sin(30°) - 3$$
$$A_x = 1.8 - 3$$
$$A_x = 1.2 \text{ kN} \leftarrow$$

$$(1.5\text{m})(3\text{kN}) + (2\text{m})(4\text{kN}) = (4\text{m})\, N_c \cos\theta$$
$$4.5 + 8 = 4 N_c \cos(30°)$$
$$\frac{12.5}{4} = N_c \cos(30°) \qquad N_c = \frac{3.125}{\cos(30°)}$$

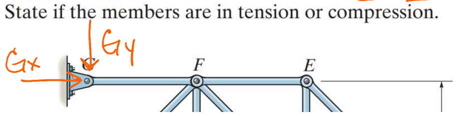
$$N_c = 3.6 \text{ kN}$$

$$\Sigma F_y = 0 \rightarrow A_y - 4 + (N_c)_y = 0$$
$$A_y = 4 - N_c \cos\theta$$
$$A_y = 4 - 3.6 \cos(30°)$$
$$A_y = 0.875 \text{ kN} \uparrow$$

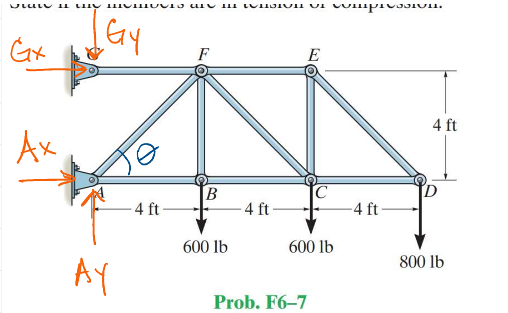
F6–7. Determine the force in members BC, CF, and FE. State if the members are in tension or compression.

$\theta = 45°$

* First solve for reaction @ supports.



State if the members are in tension or compression.



Prob. F6–7

★ First solve for reaction @ supports.

$\circlearrowleft + \Sigma M_A = 0$ neglect A_x, A_y, G_y

$$\circlearrowleft + \Sigma M_A = -4(600) - 8(600) - 12(800) - 4G_x = 0$$

$$\underbrace{-2400 - 4800 - 9600}_{} - 4G_x = 0$$

$$-16{,}800 = 4G_x$$

$$G_x = -4200 \text{ lb}$$

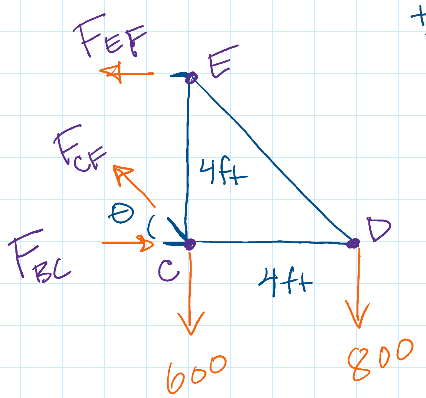
$$G_x = 4200 \text{ lb} \leftarrow$$

$\xrightarrow{+} \Sigma F_x = 0$

$$A_x = 4200 \text{ lb} \rightarrow$$

$+\uparrow \Sigma F_y = 0 \rightarrow A_y - G_y - 600 - 600 - 800 = 0$

$$A_y - G_y = 2000 \text{ lb}$$



$\circlearrowleft + \Sigma M_C = 0 \quad -4(800) + 4F_{EF} = 0$

$$\cancel{4}F_{EF} = \cancel{4}(800) \qquad F_{EF} = 800 \text{ lb (T)}$$

$+\uparrow \Sigma F_y = 0 \qquad F_{CF}\sin(45°) - 600 - 800 = 0$

$$F_{CF}\sin(45°) = 1400$$

$$F_{CF} = \frac{1400}{\sin(45°)} = 1980 \text{ lb (T)}$$

$\theta = 45°$

$\xrightarrow{+} \Sigma F_x = 0 \qquad -F_{EF} - F_{CF}\cos(45°) + F_{BC} = 0$

$$+800 + 1980\cos(45°) = F_{BC}$$

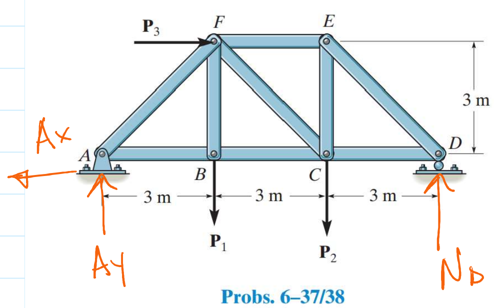
$$+800 + 1400 = F_{BC}$$

$$F_{BC} = +2200 \qquad F_{BC} = 2200 \text{ lb (C)}$$

6–38. Determine the force in members BC, ~~BE~~ FC, and EF of the truss and state if these members are in tension or compression. Set $P_1 = 6$ kN, $P_2 = 9$ kN, and $P_3 = 12$ kN.

$\Sigma M_A = 0$

of the truss and state if these members are in tension or compression. Set $P_1 = 6$ kN, $P_2 = 9$ kN, and $P_3 = 12$ kN.



Probs. 6–37/38

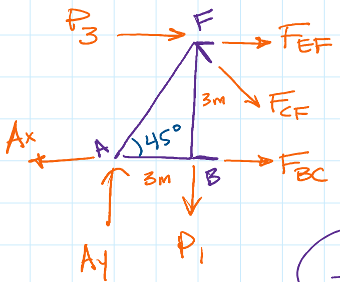
$$\Sigma M_A = 0$$

$$9 N_D = 3P_1 + 6P_2 + 3P_3$$

$$N_D = \frac{3(6) + 6(9) + 3(12)}{9}$$

$$N_D = 12 \text{ kN} \uparrow$$

$$A_x = 12 \text{kN} \leftarrow$$
$$A_y = 3 \text{ kN} \uparrow$$



$$+\circlearrowleft \Sigma M_F = 0$$

Don't account P_1, P_3, F_{EF}, F_{CF}

$$3F_{BC} - 3A_x - 3A_y = 0$$

$$F_{BC} = \frac{\cancel{3}(12+3)}{\cancel{3}} = 15 \text{ kN}$$

$$F_{BC} = 15 \text{ kN (T)}$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0$$

$$F_{BC} + F_{CF}\cos(45°) + F_{EF} + P_3 - A_x = 0$$

$$F_{EF} + F_{CF}\cos(45°) = A_x - P_3 - F_{BC}$$
$$F_{EF} + F_{CF}\cos(45°) = 12 - 12 - 15$$
$$F_{EF} + F_{CF}\cos(45°) = -15$$

$$F_{EF} + (-4.24)\cos(45°) = -15$$
$$F_{EF} - 3 = -15$$
$$F_{EF} = -12 \text{ kN}$$

$$F_{EF} = 12 \text{kN (C)}$$

$$+\uparrow \Sigma F_y = 0$$

$$A_y - P_1 - F_{CF}\sin(45°) = 0$$
$$F_{CF}\sin(45°) = A_y - P_1$$
$$F_{CF}\sin(45°) = 3 - 6$$
$$F_{CF} = \frac{-3}{\sin(45°)} = -4.24 \text{ kN}$$

$$F_{CF} = 4.24 \text{ kN (C)}$$